



Bees can count

According to research, bees can actually count with very small numbers of nerve cells in their brains. <https://digit.in/jan19-75>



Pothos ivy

Researchers have modified a common houseplant, Pothos ivy, to suck cancer-related chemicals from air. <https://digit.in/jan19-76>

In the remote Namibian desert, there are roughly circular patches of land surrounded by growth of short grass. In some locations, the grassy vegetation may be peppered with these barren patches. The local Himba, the nomadic and pastoral people, explain the phenomena as footprints of their gods. Another indigenous group of the same region, the San or the bushmen, associate the circles with spiritual and magical powers.

Despite the remote location, the circles have become a tourist attraction. Enthusiastic tour guides claim that the circles are a result of a fire breathing dragon that lives below the surface. Other esoteric explanations include meteorite bombardment and UFO activities. Scientists have no concrete explanation so far, although the circles have been studied since the 1920s.

LIFE AND DEATH

While the exact cause and purpose of the fairy circles are unknown, they have

been studied extensively, and have even been tracked by satellites over many years. Their sizes range from between 2 meters to 12 meters in diameter. The fairy circles have a life cycle of their own. At the birth of a fairy circle, the vegetation in the middle starts to die off. It usually begins as a red patch. Then there is the maturation period, where tall grass grows around the margins of the circle, and the feature becomes more defined. The circles are born almost about the same size as they will ever be, and do not grow. Even if the circles do grow, then they do so within a span of four years. It is also possible that some circles grow, while others appear fully formed. The circles typically remain in the mature state for many years. During this period, the effects of the wind hollows out the circle, forming shallow concave pits.

Some circles go through an enlargement stage during this period of stasis. This enlargement is not gradual, but rather dramatic. First, a red patch of ground appears all around the circle. Within this patch, there is a mixture

of living, dying and dead plants. At times, the red patch appears to only one side of the circle. Now, a fresh ring of tall grass starts growing around the expanded area of the circle, with everything within dying off. After a period of stability that lasts for anywhere between 30 to 60 years, the circles start dying off. The first signs of the circles dying off is the emergence of short grass and vegetation in the middle of the circles. Eventually, the circles are revegetated. The ring of tall grass remains around the circles, as well as the concavity in the middle. This means that the "ghosts" of fairy circles can be recognised even after they have died.

At times, when a fairy circle is dying, it results in the formation of a new fairy circle, that is in tangential contact with it. It is unknown if the dying of an old fairy circle in some way causes a new one to be formed nearby. In very, very rare instances, a ghost or dead fairy circle may lose the vegetation in the middle, and be born again. The number of fairy circles that are dying or being born at any time is only a fraction of

FAIRY CIRCLES

Mysterious circular patches of barren ground in Namibia have managed to confound scientists for over a century

Aditya Madanapalle | aditya@digit.in